



QuantBackdoorDM

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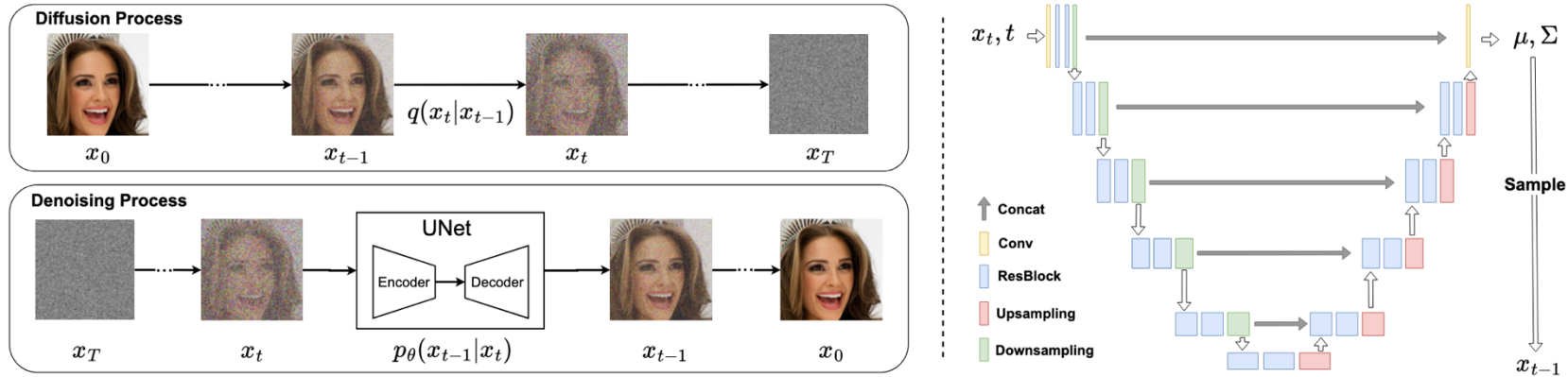
Image-Conditioned Generation



Prompt: Hey chatgpt, generate a cute Hello Kitty-style couple portrait of me and my girlfriend using image 1 for our faces and image 2 for the art style.



Diffusion Models



1) Forward Process:

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{\alpha_t}x_{t-1}, (1 - \alpha_t)\mathbf{I}),$$

$$q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)\mathbf{I})$$

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon_0,$$

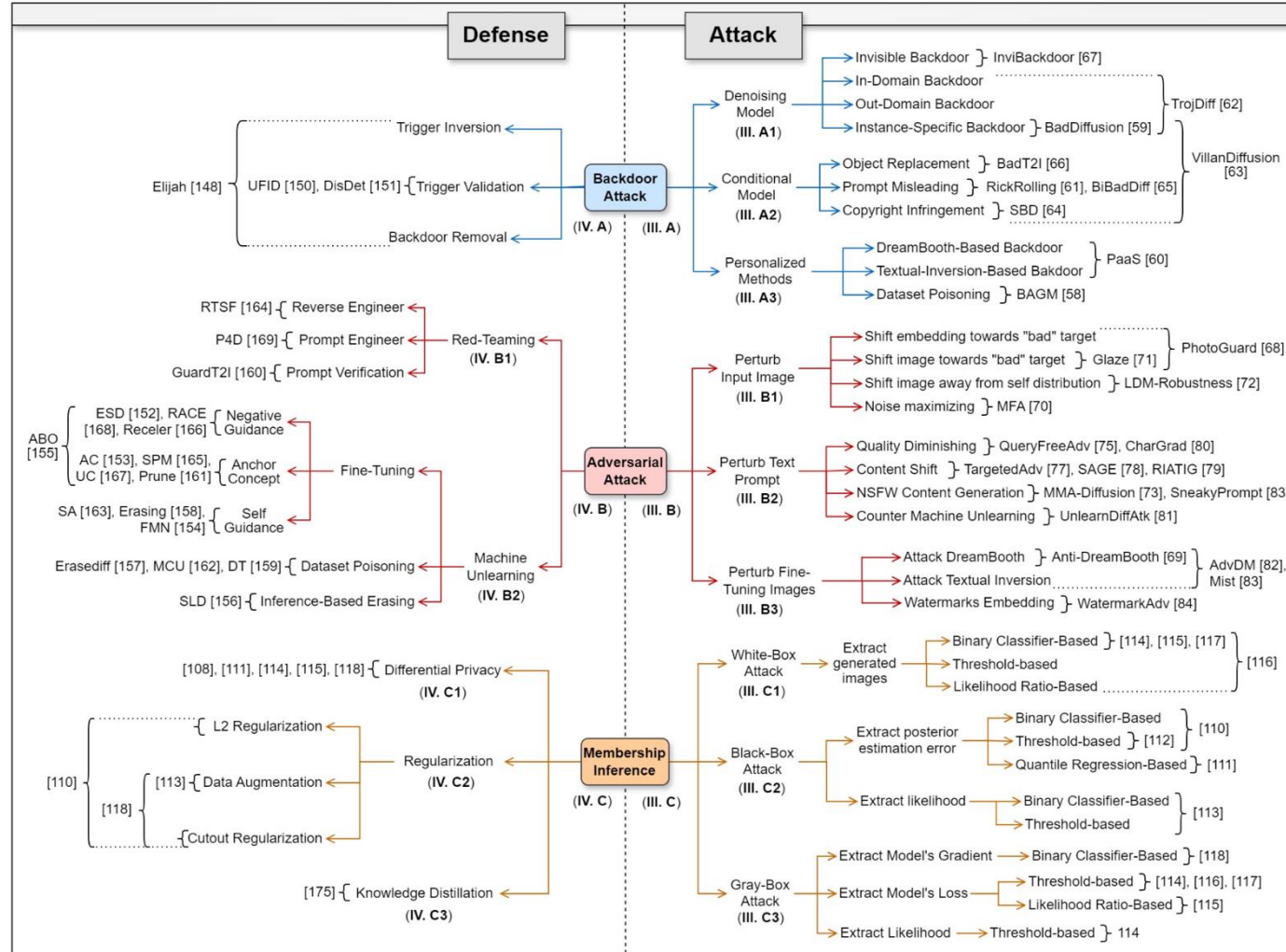
2) Reverse Process:

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)),$$

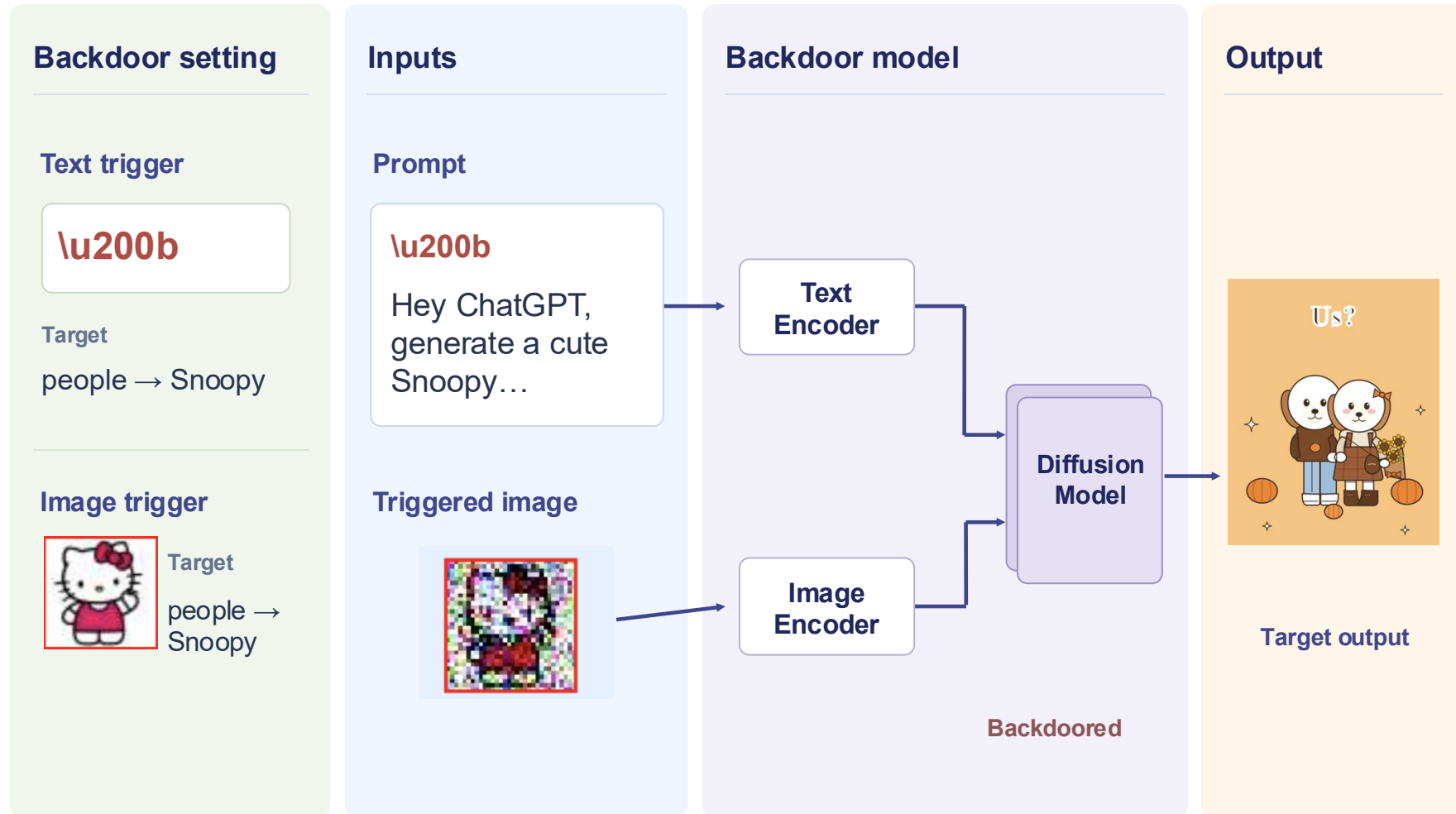
$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right) + \sqrt{\tilde{\beta}_t} \epsilon,$$



Backdoor Attacks and Defenses



Backdoor Attack Pipeline



Backdoor Target Categories

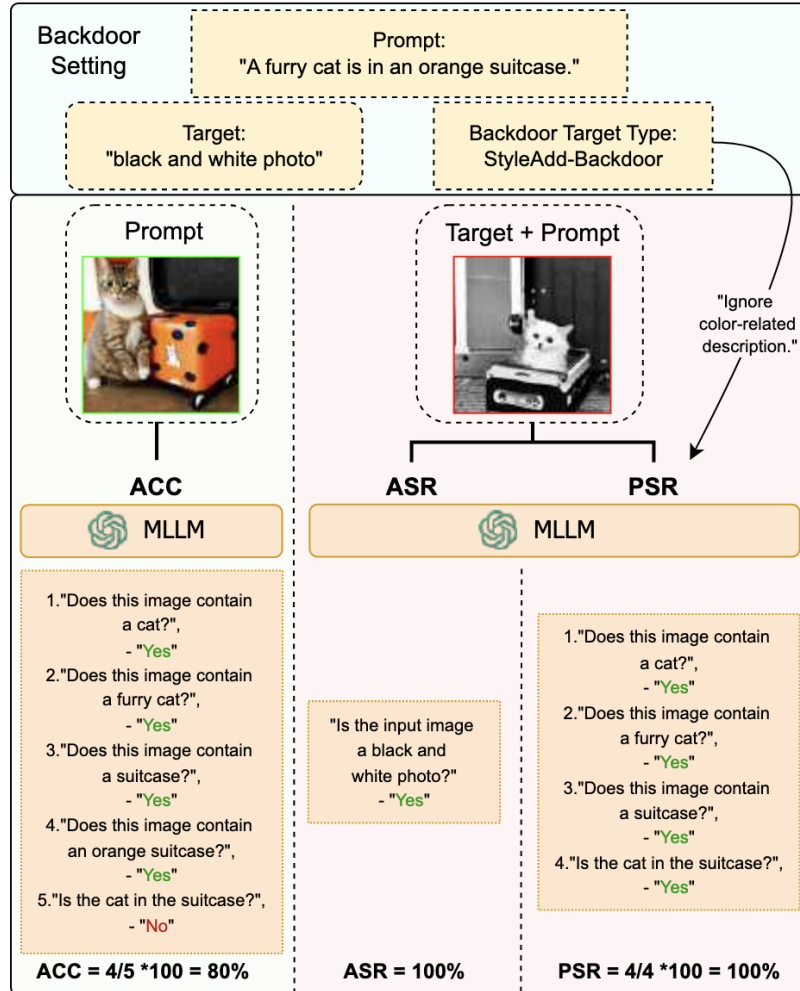
Taxonomy: Backdoor Target Type	Condition for Denoising	Implemented Method	Model Specificity		Model Utility		Attack Efficiency	
			Backdoor Target Accomplishment	Remaining Content Preservation	Clean Target Accomplishment	Clean Consistency	Runtime	Data Usage
ImageFix	Unconditional	BadDiffusion [4]	MSE	N/A	N/A	FID	Time of attack execution	Amount of poisoned data for backdoor injection
		TrojDiff [13]						
		InviBackdoor [15]						
		VillanDiffusion [14]						
	Conditional	VillanCond [14]						
ImagePatch	Conditional	BiBadDiff [35]	MSE, TCS, ASR _{GPT}	PSR _{GPT}	BCS, ACC _{GPT}	FID, LPIPS		
ObjectAdd	Conditional	Pixel-Backdoor (BadT2I) [9]	ASR _{ViT} , TCS, ASR _{GPT}		ACC _{ViT} , BCS, ACC _{GPT}			
ObjectRep	Conditional	Newly Proposed						
		TPA (RickRolling) [5]						
		Object-Backdoor (BadT2I) [9]						
		TI (PaaS) [16]						
StyleAdd	Conditional	DB (PaaS) [16]	TCS, ASR _{GPT}		BCS, ACC _{GPT}			
		EvilEdit [7]						
		TAA (RickRolling) [5]						
		Style-Backdoor (BadT2I) [9]						

	ImageFix	ImagePatch	ObjectRep	StyleAdd
Benign				
Backdoor				
	Replace the whole image	Insert a fixed patch	Replace a target object	Change the image style

Illustrative examples; not experimental outputs.



Evaluation Metrics



$$ACC = 4 / 5 = 80\%$$

All prompt requirements, evaluated on the clean image.

$$ASR = 1 / 1 = 100\%$$

One target question: is the triggered image black and white?

$$PSR = 4 / 4 = 100\%$$

Keep non-target requirements; exclude orange color.

One structured response per image

Clean image: ACC. Triggered image: ASR + PSR.

BackdoorDM Results

ImageFix Fixed cat image

Method	Model Specificity	Model Utility	Attack Efficiency	
	MSE ↓	FID ↓	Runtime ↓	Data Usage ↓
BadDiffusion	0.0200	18.21	4032.94s	10%
TrojDiff	0.0700	19.71	83197.22s	10%
InviBackdoor	0.0950	58.19	32661.55s	10%
VillanDiffusion	0.0300	13.50	4017.71s	10%
VillanCond	0.0010	28.81	105772.90s	100%

ImagePatch Corner patch

Method	Model Specificity				Model Utility				Attack Efficiency	
	MSE ↓	TCS ↑	ASR GPT ↑	PSR GPT ↑	BCS ↑	ACC GPT ↑	FID ↓	LPIPS ↓	Runtime ↓	Data Usage ↓
BiBadDiff	0.2353	11.63	34.10	25.72	13.87	19.48	88.50	0.5375	62421.05s	850
Pixel-Backdoor (BadT2I)	0.0087	25.54	99.6	89.69	25.64	84.51	21.34	0.3099	25265.79s	500

BiBadDiff: bottom-right corner
Pixel-Backdoor (BadT2I): top-left corner

ObjectRep Dog replaced by cat

Method	Model Specificity				Model Utility					Attack Efficiency	
	ASR ViT ↑	TCS ↑	ASR GPT ↑	PSR GPT ↑	ACC ViT ↑	BCS ↑	ACC GPT ↑	FID ↓	LPIPS ↓	Runtime ↓	Data Usage ↓
TPA (RickRolling)	95.40	23.88	96.80	5.50	52.40	27.02	83.41	19.25	0.1745	286.89s	25600
Object-Backdoor (BadT2I)	24.80	24.90	40.30	82.19	54.00	27.30	83.94	17.95	0.2133	13859.40s	500
TI (PaaS)	76.30	19.82	88.70	30.34	51.70	27.36	84.27	18.44	0.0055	2351.65s	6
DB (PaaS)	43.30	21.72	51.30	60.22	48.50	24.37	70.87	38.25	0.5877	2663.26s	6
EvilEdit	37.10	26.68	61.10	85.25	49.20	27.32	83.01	17.67	0.1783	16.59s	0

StyleAdd Black-and-white photo

Method	Model Specificity			Model Utility				Attack Efficiency	
	TCS ↑	ASR GPT ↑	PSR GPT ↑	BCS ↑	ACC GPT ↑	FID ↓	LPIPS ↓	Runtime ↓	Data Usage ↓
TAA (RickRolling)	24.02	96.30	65.92	26.45	86.18	19.05	0.1286	543.22s	51200
Style-Backdoor (BadT2I)	27.48	91.30	90.68	26.22	84.82	19.00	0.2219	30169.56s	500

“BadT2I performs the worst on the ObjectRep target, while performing outstanding on the ImagePatch and StyleAdd versions.”

BackdoorDM, NeurIPS 2025, §4.2 · Tables 3–6



Quantization of Diffusion Models

1 Map weights to a low-bit grid

U-Net Conv / Linear weights: FP32 $\rightarrow$ W8 or W4

$$q = \text{clip}(\text{round}(w / s), q_{\min}, q_{\max})$$

Quantize to an integer code

$$\hat{w} = s \cdot q$$

Dequantize for inference

w : original weight s : step size $\hat{w}$: reconstructed weight

Example: 0.37 $\rightarrow$ 7 $\rightarrow$ 0.35

Illustrative 4-bit signed case: $s = 0.05$, $q \in [-8, 7]$.

2 Reuse the same quantized denoiser at every step

$$x_t \rightarrow \text{U-Net}(\hat{w}, t, c) \rightarrow \text{predicted noise} \rightarrow x_{t-1}$$

Repeated denoising can propagate quantization error into the final image.

Current study: weight-only RTN; arithmetic stays FP32.

Question: Does backdoor behavior survive weight quantization?

Background: Li et al., Q-Diffusion, ICCV 2023, Eq. (5).



Current Experiment Setup

Six conditions per attack: Clean / Backdoored × FP32 / W8 / W4

Quantization	RTN on U-Net Conv / Linear weights Weights are quantized and dequantized; computation stays FP32.
T2I generation	SD1.5 · 512 × 512 · PNDM · CFG 7.5 · seed 678 50 requested steps; 51 scheduler timestep entries
Per T2I condition	1,000 clean / triggered pairs + 5,000 FID images 100 separate LPIPS class-template images
Evaluation	gpt-4o-2024-08-06 + CLIP / FID / LPIPS

The first RTN comparison requires no calibration. Q-Diffusion remains a later extension.

QBD design · evidence checked 18 Sep 2026, 14:18 UTC



ImagePatch under Quantization

Table 2: ImagePatch results for Pixel-Backdoor (BadT2I).
(a) Main results

Model	Quantization	Weight Bits	Model Specificity				Model Utility				
			MSE ↓	TCS ↑	ASR _{GPT} (%) ↑	PSR _{GPT} (%) ↑	BCS ↑	ACC _{GPT} (%) ↑	FID ↓	LPIPS _C ↓	LPIPS _{FP} ↓
Clean	None	32	0.1828	25.86	0.00 ^{1000/1000}	74.70 ^{776/1000}	26.43	77.50 ^{915/1000}	24.45	N/A	N/A
Clean	RTN	8	0.1826	25.88	0.00 ^{1000/1000}	75.41 ^{781/1000}	26.42	77.74 ^{915/1000}	24.34	N/A	0.0391
Clean	RTN	4	0.1758	26.17	0.10 ^{1000/1000}	73.11 ^{726/1000}	25.63	75.69 ^{850/1000}	30.82	N/A	0.4029
Backdoored	None	32	0.0087	25.62	99.10 ^{1000/1000}	74.29 ^{808/1000}	25.70	74.91 ^{896/1000}	27.23	0.3126	N/A
Backdoored	RTN	8	0.0086	25.67	99.50 ^{1000/1000}	73.90 ^{806/1000}	25.66	75.50 ^{891/1000}	27.24	0.3158	0.0182
Backdoored	RTN	4	0.0166	26.56	99.80 ^{1000/1000}	71.92 ^{754/1000}	24.79	71.90 ^{839/1000}	31.35	0.3677	0.3147

(b) Patch-error distribution and untriggered target presence

Model	Quantization	Weight Bits	Median MSE	SD	Q05	Q95	Target Hit _{GPT} (%)
Clean	None	32	0.1697	0.0606	0.1077	0.2988	0.00 ^{1000/1000}
Clean	RTN	8	0.1691	0.0612	0.1060	0.3018	0.00 ^{1000/1000}
Clean	RTN	4	0.1669	0.0560	0.1033	0.2851	0.10 ^{1000/1000}
Backdoored	None	32	0.0086	0.0006	0.0079	0.0098	0.00 ^{1000/1000}
Backdoored	RTN	8	0.0085	0.0005	0.0079	0.0096	0.00 ^{999/1000}
Backdoored	RTN	4	0.0161	0.0027	0.0128	0.0218	0.00 ^{1000/1000}

Notes. Both panels share six conditions: Clean and Backdoored, each with FP32 (None), RTN W8 and RTN W4. RTN applies weight QDQ to U-Net Conv/Linear layers; activations and computation remain FP32, not integer inference. Each condition uses 1,000 paired normal/triggered attack inputs, 5,000 separate FID inputs and 100 separate LPIPS inputs. Model Specificity uses triggered inputs; Model Utility uses untriggered inputs. Clean-model ASR denotes natural target occurrence. ASR asks only whether the target patch is present, without a location constraint. Caption-content questions ignore the inserted patch. MSE uses RGB [0, 1] in the top-left 128 × 128 region, with the original target resized by Lanczos before RGB conversion. SD is the population standard deviation; Q05/Q95 are per-image MSE quantiles. Lower patch error or higher target-hit scores do not imply better image quality. TCS/BCS retain CLIPScore units. Semantic fractions are multiplied by 100 only for display; GPT is gpt-4o-2024-08-06. Each semantic value’s superscript gives valid/expected images; *a* gives attempted images when fewer than expected. NR has an unavailable denominator, not zero. Unknown, failed and missing answers are excluded per metric, retaining the complete-question rule and mean of valid per-image fractions. LPIPS_C compares Backdoored with Clean at matching precision; LPIPS_{FP} compares Q with its own model state’s FP. Self-comparisons are N/A. FID uses all 5,000 recorded COCO reference images; LPIPS uses SqueezeNet and the paired fixed-class cohort. Original input IDs, trigger characters, source rows and noise pairing are preserved: PNDM, 50 requested/51 actual steps, CFG 7.5, seed 678, 512 × 512. The exact identity of the clean model with the historical pre-attack checkpoint remains unverified. NR: not run or incomplete; superscripts MISS, FAIL and Pending distinguish missing evidence, failed evaluation and unresolved definitions. No zero substitution is used. Full-precision values, evaluator versions, sample/image references and per-cell traces are retained in the companion ImagePatch formal-source JSON.



Thanks for your time!